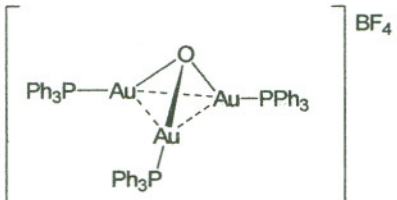
Organic Cume
9/24/04

Gold-Catalyzed Organic Reactions

1) Gold is often used as a Lewis acid in certain organic transformations. Would you expect compound **1** or compound **2** to be the stronger Lewis acid? Explain your choice.

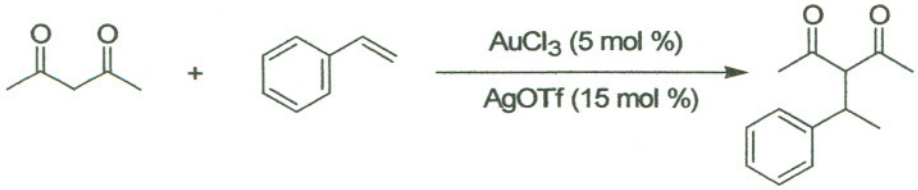


1

[(PPh$_3$)Au][BF$_4$]

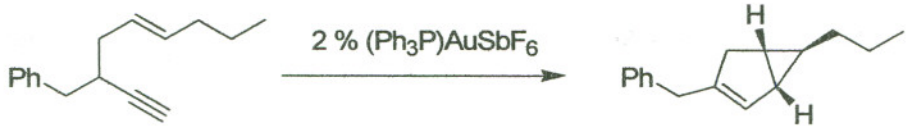
2

2) This year, Li and co-workers reported the gold-mediated transformation described in eq 1 (*JACS* **2004**, 6884). Provide a reasonable step-by-step mechanism for a catalytic cycle and clearly note the oxidation state of the gold for each intermediate.



(1)

3) Toste recently reported the catalytic isomerization of 1,5-enynes to bicyclo[3.1.0.]hexenes mediated by gold (eq 2, *JACS* **2004**, 10858). Provide a reasonable step-by-step mechanism for a catalytic cycle and clearly note the oxidation state of the gold for each intermediate.



(2)